Develop a movie recommendation model using the scikit-learn library in python.

In [1]:
```
from sklearn.metrics.pairwise import cosine_similarity
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

In [2]:
```
df = pd.read_csv("https://raw.githubusercontent.com/rashida048/Some-
```

In [3]:
```
df.head()
```

Out[3]:

	index	budget	genres	homepage	id	keywords	original_langu
0	0	237000000	Action Adventure Fantasy Science Fiction	http://www.avatarmovie.com/	19995	culture clash future space war space colony so...	
1	1	300000000	Adventure Fantasy Action	http://disney.go.com/ disneypictures/pirates/	285	ocean drug abuse exotic island east india trad...	
2	2	245000000	Action Adventure Crime	http:// www.sonypictures.com/ movies/spectre/	206647	spy based on novel secret agent sequel mi6	
3	3	250000000	Action Crime Drama Thriller	http:// www.thedarkknightrises.com/	49026	dc comics crime fighter terrorist secret ident...	
4	4	260000000	Action Adventure Science Fiction	http://movies.disney.com/ john-carter	49529	based on novel mars medallion space travel pri...	

5 rows × 24 columns

In [4]:
```
features = ['keywords','cast','genres','director']
```

In [5]:
```
def combine_features(row):
    return row['keywords']+" "+row['cast']+" "+row['genres']+" "+row
```

In [6]:
```
for feature in features:
    df[feature] = df[feature].fillna('')

df["combined_features"] = df.apply(combine_features,axis=1)
```

In [7]:
```
cv = CountVectorizer()
count_matrix = cv.fit_transform(df["combined_features"])
```

In [8]:
```
cosine_sim = cosine_similarity(count_matrix)
```

In [9]:
```
def get_title_from_index(index):
    return df[df.index == index]["title"].values[0]
def get_index_from_title(title):
    return df[df.title == title]["index"].values[0]
```

In [10]:
```
movie_user_likes = "Avatar"
movie_index = get_index_from_title(movie_user_likes)
similar_movies = list(enumerate(cosine_sim[movie_index]))
```

In [11]:
```
sorted_similar_movies = sorted(similar_movies,key=lambda x:x[1],reve
```

In [12]:
```
i=0
print("Top 5 similar movies to "+movie_user_likes+" are:\n")
for element in sorted_similar_movies:
    print(get_title_from_index(element[0]))
    i=i+1
    if i>5:
        break
```

```
Top 5 similar movies to Avatar are:

Guardians of the Galaxy
Aliens
Star Wars: Clone Wars: Volume 1
Star Trek Into Darkness
Star Trek Beyond
Alien
```

In [ ]: